

DSA 8301: Statistical Inference for Big Data

Lecture Notes

Topic 5: Maximum Likelihood Estimation

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5 Maximum Likelihood Estimation

5.1 Introduction

Maximum Likelihood Estimation (MLE) is one of the most important and widely used methods for estimating unknown parameters in statistics.

Basic Idea The central idea of MLE is:

Choose the value of the parameter that makes the observed data most likely to occur.

In other words, among all possible values of the parameter, we select the one that best explains the data we have observed.

Statistical Setting Suppose we observe a random sample:

$$X_1, X_2, \dots, X_n,$$

which are assumed to be:

- **independent**, and
- **identically distributed (i.i.d.)**.

Each observation comes from a distribution with:

$$f(x; \theta),$$

where:

- $f(x; \theta)$ is either:
 - a **probability density function (pdf)** (for continuous data), or
 - a **probability mass function (pmf)** (for discrete data),
- θ is an **unknown parameter** that we want to estimate,
- Θ is the **parameter space** (all possible values of θ).

Goal Our objective is to use the observed data:

$$x_1, x_2, \dots, x_n,$$

to find a value $\hat{\theta}$ that best describes the underlying distribution.

Terminology

- The function that measures how likely the data are for different values of θ is called the **likelihood function**.
- The value of θ that maximizes this function is called the **maximum likelihood estimator (MLE)**.

5.2 Likelihood and Log-Likelihood Functions

Definition 5.1 (Likelihood Function). *Let $X_1, X_2, \dots, X_n$ be an i.i.d. sample from a distribution with density (or pmf) $f(x; \theta)$.*

*The **likelihood function** is defined as:*

$$L(\theta; x_1, \dots, x_n) = \prod_{i=1}^n f(x_i; \theta),$$

where $x_1, \dots, x_n$ are the observed data.

5.2.1 Derivation of the Likelihood Function

Step 1: Joint Distribution Since the observations are independent, the joint density (or pmf) is:

$$f(x_1, \dots, x_n; \theta) = \prod_{i=1}^n f(x_i; \theta).$$

Step 2: View as a Function of θ

- Normally, we think of $f(x; \theta)$ as a function of x .
- In likelihood theory, we fix the observed data and treat θ as the variable.

Thus, the same expression is now viewed as a function of θ , called the **likelihood function**.

5.2.2 Interpretation

- The likelihood function measures how well different values of θ explain the observed data.
- Larger values of $L(\theta)$ indicate that θ is more consistent with the data.

Important Remark

- The likelihood is **not** a probability distribution in θ .
- It does not have to integrate (or sum) to 1.
- It is simply a tool for comparing different parameter values.

5.2.3 Example (Discrete Case)

Suppose $X_i \sim \text{Bernoulli}(\theta)$, where:

$$P(X_i = x_i) = \theta^{x_i}(1 - \theta)^{1-x_i}.$$

Then the likelihood is:

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n \theta^{x_i}(1 - \theta)^{1-x_i}. \\ &= \theta^{\sum x_i}(1 - \theta)^{n - \sum x_i}. \end{aligned}$$

5.2.4 Log-Likelihood Function

Definition 5.2 (Log-Likelihood). *The **log-likelihood function** is defined as:*

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i; \theta).$$

5.2.5 Derivation

Starting from:

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta),$$

take logarithms:

$$\ell(\theta) = \log \left(\prod_{i=1}^n f(x_i; \theta) \right).$$

Using the log property:

$$\log(ab) = \log a + \log b,$$

we obtain:

$$\ell(\theta) = \sum_{i=1}^n \log f(x_i; \theta).$$

5.2.6 Why Use the Log-Likelihood?

1. Simplifies Calculations

- Products are often complicated to differentiate.
- Logarithms convert products into sums, which are easier to handle.

2. Easier Differentiation

- Differentiating sums is simpler than differentiating products.
- This is very useful when solving for the MLE.

3. Numerical Stability

- Likelihoods can involve very small numbers (products of probabilities).
- Taking logs avoids numerical underflow in computations.

4. Same Maximizer

- The logarithm is a strictly increasing function.
- Therefore:

$$\arg \max_{\theta} L(\theta) = \arg \max_{\theta} \ell(\theta).$$

5.3 Derivation of Maximum Likelihood Estimators

Definition 5.3 (Maximum Likelihood Estimator). *The **maximum likelihood estimator (MLE)** of a parameter θ is the value that maximizes the likelihood function:*

$$\hat{\theta} = \arg \max_{\theta \in \Theta} L(\theta) = \arg \max_{\theta \in \Theta} \ell(\theta),$$

where $L(\theta)$ is the likelihood and $\ell(\theta)$ is the log-likelihood.

The notation:

$$\arg \max_{\theta \in \Theta} L(\theta)$$

means:

“the value of θ (from the set Θ) that maximizes the function $L(\theta)$.”

Breaking it down

- θ is the parameter we want to estimate.
- Θ is the set of all possible values of θ (called the parameter space).
- $\max$ refers to the maximum value of a function.
- $\arg \max$ refers to the **argument** (input value) at which the maximum occurs.

Important Distinction

- $\max L(\theta)$ gives the **maximum value** of the likelihood.
- $\arg \max L(\theta)$ gives the **value of θ** that achieves this maximum.

5.3.1 Procedure for Finding the MLE

Step 1: Write the Likelihood Function

- Start with the joint distribution of the sample:

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta).$$

- This expresses the likelihood of observing the data given θ .

Step 2: Take Logarithms

- Compute the log-likelihood:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i; \theta).$$

- This simplifies calculations by turning products into sums.

Step 3: Differentiate

- Take the derivative with respect to θ :

$$\frac{d}{d\theta} \ell(\theta).$$

- This measures how the log-likelihood changes as θ changes.

Step 4: Solve the Likelihood Equation

- Set the derivative equal to zero:

$$\frac{d}{d\theta} \ell(\theta) = 0.$$

- Solve for θ to obtain candidate values for the MLE.

Step 5: Check Second-Order Conditions

- Compute the second derivative:

$$\frac{d^2}{d\theta^2} \ell(\theta).$$

- If:

$$\frac{d^2}{d\theta^2} \ell(\theta) < 0,$$

then the solution corresponds to a maximum.

Example 5.3.1**Bernoulli Case**

Let $X_i \sim \text{Bernoulli}(\theta)$.

Step 1: Likelihood

$$L(\theta) = \theta^{\sum x_i} (1 - \theta)^{n - \sum x_i}.$$

Step 2: Log-Likelihood

$$\ell(\theta) = \sum x_i \log \theta + (n - \sum x_i) \log(1 - \theta).$$

Step 3: Differentiate

$$\frac{d\ell}{d\theta} = \frac{\sum x_i}{\theta} - \frac{n - \sum x_i}{1 - \theta}.$$

Step 4: Solve Set equal to zero:

$$\frac{\sum x_i}{\theta} = \frac{n - \sum x_i}{1 - \theta}.$$

Solve:

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{X}.$$

Conclusion

- The MLE of θ is the sample mean.

—

5.3.2 Example 1: Bernoulli Distribution

Let $X_i \sim \text{Bernoulli}(p)$.

$$L(p) = \prod_{i=1}^n p^{x_i} (1 - p)^{1 - x_i} = p^{\sum x_i} (1 - p)^{n - \sum x_i}.$$

$$\ell(p) = \sum x_i \log p + (n - \sum x_i) \log(1 - p).$$

Differentiate:

$$\frac{d\ell}{dp} = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1 - p}.$$

Set to zero:

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}.$$

Example 5.3.2**Normal Mean (Variance Known)**

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables such that:

$$X_i \sim N(\mu, \sigma^2),$$

where σ^2 is known and μ is the unknown parameter.

Step 1: Write the Density Function The normal density is:

$$f(x_i; \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right).$$

Step 2: Write the Likelihood Function Since the observations are independent:

$$L(\mu) = \prod_{i=1}^n f(x_i; \mu).$$

Substitute the density:

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right).$$

Step 3: Simplify the Product

- The constant terms multiply:

$$\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n$$

- The exponentials combine:

$$\exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right)$$

Thus:

$$L(\mu) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2\right).$$

Step 4: Take Logarithms

$$\ell(\mu) = \log L(\mu)$$

$$= -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Step 5: Differentiate the Log-Likelihood We differentiate term by term.

- The first term is constant in μ , so its derivative is 0.
- Focus on:

$$-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2.$$

Differentiate inside the sum:

$$\frac{d}{d\mu}(x_i - \mu)^2 = -2(x_i - \mu).$$

Thus:

$$\frac{d\ell}{d\mu} = -\frac{1}{2\sigma^2} \sum [-2(x_i - \mu)].$$

Step 6: Simplify

$$= \frac{1}{\sigma^2} \sum (x_i - \mu).$$

Step 7: Set the Derivative to Zero

$$\frac{1}{\sigma^2} \sum (x_i - \mu) = 0.$$

Multiply both sides by σ^2 :

$$\sum (x_i - \mu) = 0.$$

Step 8: Solve for μ

$$\sum x_i - n\mu = 0.$$

$$n\mu = \sum x_i.$$

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{X}.$$

Step 9: Second Derivative Check

$$\frac{d^2\ell}{d\mu^2} = -\frac{n}{\sigma^2} < 0.$$

Thus, the solution corresponds to a maximum.

Final Result

$$\boxed{\hat{\mu} = \bar{X}}$$

5.4 Fisher Information

Fisher Information is a key concept in statistical inference. It measures how much information a random variable carries about an unknown parameter θ .

5.4.1 Score Function

Definition 5.4 (Score Function). *The **score function** is defined as:*

$$U(X; \theta) = \frac{\partial}{\partial \theta} \log f(X; \theta).$$

5.4.2 Interpretation of the Score Function

- The score function measures how sensitive the log-likelihood is to changes in θ .
- It tells us how much the data “push” the estimate of θ .
- A large value of $U(X; \theta)$ means that small changes in θ greatly affect the likelihood.

Key Property

$$\mathbb{E}_\theta[U(X; \theta)] = 0.$$

Meaning:

- On average, the score function is zero at the true parameter value.
- This reflects that the likelihood is maximized at the true parameter.

5.4.3 Fisher Information

When we estimate a parameter θ from data, not all samples are equally “informative.” Fisher information quantifies how much information an observable random variable carries about an unknown parameter. It is central to the Cramér–Rao lower bound (CRLB), which limits the precision of unbiased estimators.

Definition 5.5 (One variable). *Let X be a random variable with density (or pmf) $f(x; \theta)$ depending on parameter θ . The **score function** (derivative of the log-likelihood with respect to the parameter) is*

$$U(X; \theta) = \frac{\partial}{\partial \theta} \log f(X; \theta).$$

*The **Fisher information** is defined as*

$$I(\theta) = \mathbb{E}_\theta[U(X; \theta)^2] = \mathbb{E}_\theta \left[\left(\frac{\partial}{\partial \theta} \log f(X; \theta) \right)^2 \right].$$

Interpretation

- $I(\theta)$ measures the *curvature* of the log-likelihood at the true parameter value.
 - A flatter log-likelihood curve means less information (estimation is harder).
 - A sharper curve means more information (estimation is easier).
- High Fisher information $\Rightarrow$ data are very informative about θ , so estimators can be more precise.
- Low Fisher information $\Rightarrow$ data carry little information, so estimation must be imprecise.

Example 5.4.1

Normal Mean (Variance Known).] Suppose $X \sim N(\mu, \sigma^2)$ with σ^2 known.

$$\log f(x; \mu) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}.$$

$$\frac{\partial}{\partial \mu} \log f(x; \mu) = \frac{x - \mu}{\sigma^2}.$$

$$I(\mu) = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma^2} \right)^2 \right].$$

We want to evaluate

$$\mathbb{E} \left[\left(\frac{X - \mu}{\sigma^2} \right)^2 \right],$$

where $X \sim N(\mu, \sigma^2)$ with μ unknown and σ^2 known.

Since σ^2 is not random,

$$\mathbb{E} \left[\left(\frac{X - \mu}{\sigma^2} \right)^2 \right] = \frac{1}{\sigma^4} \mathbb{E}[(X - \mu)^2].$$

By definition of variance,

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

For $X \sim N(\mu, \sigma^2)$,

$$\mathbb{E}[(X - \mu)^2] = \sigma^2.$$

Hence

$$\frac{1}{\sigma^4} \mathbb{E}[(X - \mu)^2] = \frac{1}{\sigma^4} \cdot \sigma^2 = \frac{1}{\sigma^2}.$$

Therefore,

$$\mathbb{E} \left[\left(\frac{X - \mu}{\sigma^2} \right)^2 \right] = \frac{1}{\sigma^2}.$$

This shows that the Fisher information for a single Normal observation with mean μ (and known variance σ^2) is

$$I(\mu) = \frac{1}{\sigma^2}.$$

So n observations give $I_n(\mu) = \frac{n}{\sigma^2}$.

Example 5.4.2

Let $X \sim \text{Bernoulli}(p)$ with $0 < p < 1$. Its pmf is

$$f(x; p) = p^x (1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

$$\ell(p; x) = \log f(x; p) = x \log p + (1 - x) \log(1 - p).$$

$$U(x; p) \equiv \frac{\partial}{\partial p} \ell(p; x) = \frac{x}{p} - \frac{1-x}{1-p}.$$

Check that the score has mean zero

$$\mathbb{E}_p[U(X; p)] = \mathbb{E}\left[\frac{X}{p} - \frac{1-X}{1-p}\right] = \frac{\mathbb{E}[X]}{p} - \frac{1 - \mathbb{E}[X]}{1-p} = \frac{p}{p} - \frac{1-p}{1-p} = 1 - 1 = 0.$$

By definition (one observation),

$$I(p) = \mathbb{E}_p[U(X; p)^2] = \sum_{x \in \{0, 1\}} U(x; p)^2 f(x; p).$$

Evaluate $U(x; p)$ at $x = 1$ and $x = 0$:

$$U(1; p) = \frac{1}{p} - 0 = \frac{1}{p}, \quad U(0; p) = 0 - \frac{1}{1-p} = -\frac{1}{1-p}.$$

Hence

$$I(p) = \left(\frac{1}{p}\right)^2 f(1; p) + \left(-\frac{1}{1-p}\right)^2 f(0; p) = \frac{1}{p^2} p + \frac{1}{(1-p)^2} (1-p) = \frac{1}{p} + \frac{1}{1-p}.$$

Combine the terms:

$$I(p) = \frac{1-p+p}{p(1-p)} = \frac{1}{p(1-p)}.$$

Additivity for n i.i.d. trials. For $X_1, \dots, X_n$ i.i.d. $\text{Bernoulli}(p)$, the sample information is

$$I_n(p) = n I(p) = \frac{n}{p(1-p)}.$$

5.4.4 Understanding the Definition

First Expression

$$I(\theta) = \mathbb{E}[U(X; \theta)^2]$$

- This is the **variance of the score function**.
- It measures how much the score fluctuates.

Second Expression

$$I(\theta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f(X; \theta) \right]$$

- This measures the **curvature** of the log-likelihood.
- A steeper curve (more curvature) means more information.

Illustration 5.4.1

Why Are the Two Expressions Equal?

We want to show that:

$$I(\theta) = \mathbb{E}[U(X; \theta)^2] = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f(X; \theta) \right].$$

Step 1: Define the Score Function

$$U(X; \theta) = \frac{\partial}{\partial \theta} \log f(X; \theta).$$

Step 2: Key Property of the Score Function Start with:

$$\mathbb{E}[U(X; \theta)] = \int \frac{\partial}{\partial \theta} \log f(x; \theta) f(x; \theta) dx.$$

Using:

$$\frac{\partial}{\partial \theta} \log f(x; \theta) = \frac{1}{f(x; \theta)} \frac{\partial}{\partial \theta} f(x; \theta),$$

we get:

$$\mathbb{E}[U(X; \theta)] = \int \frac{\partial}{\partial \theta} f(x; \theta) dx.$$

Step 3: Differentiate Under the Integral

$$\int \frac{\partial}{\partial \theta} f(x; \theta) dx = \frac{\partial}{\partial \theta} \int f(x; \theta) dx.$$

But:

$$\int f(x; \theta) dx = 1,$$

so:

$$\frac{\partial}{\partial \theta} (1) = 0.$$

Thus:

$$\mathbb{E}[U(X; \theta)] = 0.$$

Step 4: Differentiate Again Differentiate both sides with respect to θ :

$$\frac{d}{d\theta} \mathbb{E}[U(X; \theta)] = 0.$$

Step 5: Apply Product Rule

$$\frac{d}{d\theta} \mathbb{E}[U(X; \theta)] = \frac{d}{d\theta} \int U(x; \theta) f(x; \theta) dx.$$

Differentiate inside:

$$= \int \left[\frac{\partial U(x; \theta)}{\partial \theta} f(x; \theta) + U(x; \theta) \frac{\partial f(x; \theta)}{\partial \theta} \right] dx.$$

Step 6: Substitute for $\frac{\partial f}{\partial \theta}$ Recall:

$$U(X; \theta) = \frac{\partial}{\partial \theta} \log f(X; \theta) = \frac{1}{f(x; \theta)} \frac{\partial}{\partial \theta} f(x; \theta),$$

$$\frac{\partial f(x; \theta)}{\partial \theta} = f(x; \theta) U(x; \theta).$$

So:

$$\int \left[\frac{\partial U(x; \theta)}{\partial \theta} f(x; \theta) + U(x; \theta)^2 f(x; \theta) \right] dx = 0.$$

Step 7: Write in Expectation Form

$$\mathbb{E} \left[\frac{\partial U(X; \theta)}{\partial \theta} \right] + \mathbb{E}[U(X; \theta)^2] = 0.$$

Step 8: Rearrange

$$\mathbb{E}[U(X; \theta)^2] = -\mathbb{E} \left[\frac{\partial U(X; \theta)}{\partial \theta} \right].$$

Step 9: Recognize Second Derivative

$$\frac{\partial U(X; \theta)}{\partial \theta} = \frac{\partial^2}{\partial \theta^2} \log f(X; \theta).$$

Thus:

$$\mathbb{E}[U(X; \theta)^2] = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f(X; \theta) \right].$$

Final Result

$$I(\theta) = \mathbb{E}[U(X; \theta)^2] = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f(X; \theta) \right]$$

5.4.5 Intuition

- If the likelihood function is very sharp around the maximum:
 - Small changes in θ greatly reduce likelihood.

- This means we have **a lot of information**.
- If the likelihood is flat:
 - Many values of θ are plausible.
 - This means **low information**.

Example 5.4.3

Worked Example using both expression

Let $X \sim N(\mu, \sigma^2)$, where σ^2 is known. We compute Fisher Information using both definitions and show they are equal.

Step 1: Write the Density

$$f(x; \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

Step 2: Log-Likelihood

$$\log f(x; \mu) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x - \mu)^2}{2\sigma^2}.$$

Method 1: Using $I(\mu) = \mathbb{E}[U(X; \mu)^2]$

Step 3: Score Function

$$U(X; \mu) = \frac{\partial}{\partial \mu} \log f(x; \mu) = \frac{x - \mu}{\sigma^2}.$$

Step 4: Square the Score

$$U(X; \mu)^2 = \frac{(X - \mu)^2}{\sigma^4}.$$

Step 5: Take Expectation

$$I(\mu) = \mathbb{E}\left[\frac{(X - \mu)^2}{\sigma^4}\right] = \frac{1}{\sigma^4} \mathbb{E}[(X - \mu)^2].$$

Since:

$$\mathbb{E}[(X - \mu)^2] = \sigma^2,$$

we obtain:

$$I(\mu) = \frac{1}{\sigma^2}.$$

Method 2: Using $I(\mu) = -\mathbb{E}\left[\frac{\partial^2}{\partial \mu^2} \log f(X; \mu)\right]$

Step 6: First Derivative

$$\frac{\partial}{\partial \mu} \log f(x; \mu) = \frac{x - \mu}{\sigma^2}.$$

Step 7: Second Derivative

$$\frac{\partial^2}{\partial \mu^2} \log f(x; \mu) = -\frac{1}{\sigma^2}.$$

Step 8: Take Expectation

$$I(\mu) = -\mathbb{E} \left[-\frac{1}{\sigma^2} \right] = \frac{1}{\sigma^2}.$$

Conclusion

$$I(\mu) = \mathbb{E}[U(X; \mu)^2] = -\mathbb{E} \left[\frac{\partial^2}{\partial \mu^2} \log f(X; \mu) \right] = \frac{1}{\sigma^2}$$

Interpretation

- Both methods give the same Fisher Information.
- This confirms the theoretical result.
- Larger σ^2 (more variability) $\Rightarrow$ less information.
- Smaller $\sigma^2 \Rightarrow$ more precise estimation.

5.4.6 Relationship Between Fisher Information and the Likelihood Function

Fisher Information is closely related to the likelihood function and plays a central role in parameter estimation.

Step 1: Likelihood Function Given a sample $X_1, \dots, X_n$:

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta).$$

Step 2: Log-Likelihood

$$\ell(\theta) = \log L(\theta).$$

Step 3: Score Function

$$U(\theta) = \frac{d}{d\theta} \ell(\theta).$$

- The score function measures how sensitive the log-likelihood is to changes in θ .

Step 4: Fisher Information

$$I(\theta) = -\mathbb{E} \left[\frac{d^2}{d\theta^2} \ell(\theta) \right].$$

- Fisher Information measures the **curvature** (or steepness) of the log-likelihood function.

5.4.7 Interpretation in Terms of the Likelihood Function

- If the likelihood function is **steep and sharply peaked**:
 - The parameter is well identified.
 - Fisher Information is **large**.
- If the likelihood function is **flat**:
 - Many values of θ are plausible.
 - Fisher Information is **small**.

Example 5.4.4

5.4.8 Worked Example: Bernoulli Distribution

Let $X_1, X_2, \dots, X_n \sim \text{Bernoulli}(\theta)$, where:

$$P(X_i = x_i) = \theta^{x_i}(1 - \theta)^{1-x_i}, \quad 0 < \theta < 1.$$

Suppose we observe data with $\sum_{i=1}^n x_i = k$ successes.

Step 1: Likelihood Function

$$L(\theta) = \prod_{i=1}^n \theta^{x_i}(1 - \theta)^{1-x_i} = \theta^k(1 - \theta)^{n-k}.$$

Interpretation

- The likelihood shows how plausible different values of θ are.
- If k is large, larger values of θ will give higher likelihood.

Step 2: Log-Likelihood

$$\ell(\theta) = \log L(\theta) = k \log \theta + (n - k) \log(1 - \theta).$$

Step 3: Score Function

$$U(\theta) = \frac{d}{d\theta} \ell(\theta) = \frac{k}{\theta} - \frac{n - k}{1 - \theta}.$$

Interpretation

- The score measures how the likelihood changes as θ changes.
- Setting $U(\theta) = 0$ gives the MLE.

Step 4: Second Derivative

$$\frac{d^2}{d\theta^2} \ell(\theta) = -\frac{k}{\theta^2} - \frac{n-k}{(1-\theta)^2}.$$

Step 5: Fisher Information

$$I(\theta) = -\mathbb{E} \left[\frac{d^2}{d\theta^2} \ell(\theta) \right].$$

Take expectations:

$$\mathbb{E}[k] = n\theta, \quad \mathbb{E}[n-k] = n(1-\theta).$$

5.4.9 Deriving $\mathbb{E}[k] = n\theta$ and $\mathbb{E}[n-k] = n(1-\theta)$

Let:

$$X_1, X_2, \dots, X_n \sim \text{Bernoulli}(\theta),$$

where:

$$P(X_i = 1) = \theta, \quad P(X_i = 0) = 1 - \theta.$$

Define:

$$k = \sum_{i=1}^n X_i.$$

Step 1: Expectation of a Bernoulli Random Variable For each X_i :

$$\mathbb{E}[X_i] = 1 \cdot \theta + 0 \cdot (1 - \theta) = \theta.$$

Step 2: Use Linearity of Expectation

$$\mathbb{E}[k] = \mathbb{E} \left[\sum_{i=1}^n X_i \right].$$

Using linearity:

$$= \sum_{i=1}^n \mathbb{E}[X_i].$$

Since all X_i have the same expectation:

$$= \sum_{i=1}^n \theta = n\theta.$$

Step 3: Expectation of $n - k$

$$n - k = n - \sum_{i=1}^n X_i.$$

Take expectation:

$$\mathbb{E}[n - k] = \mathbb{E}[n] - \mathbb{E}[k].$$

$$= n - n\theta.$$

$$= n(1 - \theta).$$

Thus:

$$\begin{aligned} I(\theta) &= \frac{n\theta}{\theta^2} + \frac{n(1 - \theta)}{(1 - \theta)^2} = \frac{n}{\theta} + \frac{n}{1 - \theta}. \\ &= \frac{n}{\theta(1 - \theta)}. \end{aligned}$$

Final Result

$$I(\theta) = \frac{n}{\theta(1 - \theta)}$$

Relationship with Likelihood Shape

- When θ is near 0.5:
 - The likelihood is less sharp.
 - Fisher Information is smaller.
- When θ is near 0 or 1:
 - The likelihood becomes more peaked.
 - Fisher Information is larger.

5.5 Asymptotic Distribution of the Maximum Likelihood Estimator

5.5.1 Introduction

In many practical situations, it is difficult to determine the exact distribution of the maximum likelihood estimator (MLE).

However, when the sample size n is large, the MLE has a simple and useful **approximate distribution**. This is known as its **asymptotic distribution**.

Key Idea: As the sample size increases, the MLE behaves approximately like a normal (Gaussian) random variable.

5.6 Asymptotic Normality of the MLE

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with density $f(x; \theta)$, and let $\hat{\theta}$ be the maximum likelihood estimator (MLE) of θ .

Main Result Under suitable regularity conditions:

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N\left(0, \frac{1}{I(\theta)}\right).$$

Equivalently,

$$\hat{\theta} \approx N\left(\theta, \frac{1}{nI(\theta)}\right).$$

Illustration 5.6.1

Equivalence of the Two Forms

We start with the asymptotic result:

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N\left(0, \frac{1}{I(\theta)}\right).$$

Step 1: Define a New Variable Let:

$$Z_n = \sqrt{n}(\hat{\theta} - \theta).$$

Then:

$$Z_n \xrightarrow{d} N\left(0, \frac{1}{I(\theta)}\right).$$

Step 2: Express $\hat{\theta}$ in Terms of Z_n

$$\hat{\theta} - \theta = \frac{Z_n}{\sqrt{n}}.$$

Thus:

$$\hat{\theta} = \theta + \frac{Z_n}{\sqrt{n}}.$$

Step 3: Use Properties of the Normal Distribution If:

$$Z_n \xrightarrow{d} N\left(0, \frac{1}{I(\theta)}\right),$$

then for large n , we approximate:

$$Z_n \approx N\left(0, \frac{1}{I(\theta)}\right).$$

Step 4: Scale the Distribution If:

$$Z \sim N(0, \sigma^2),$$

then:

$$\frac{Z}{\sqrt{n}} \sim N\left(0, \frac{\sigma^2}{n}\right).$$

Apply this with $\sigma^2 = \frac{1}{I(\theta)}$:

$$\frac{Z_n}{\sqrt{n}} \approx N\left(0, \frac{1}{nI(\theta)}\right).$$

Step 5: Add the Constant θ If:

$$Y \sim N(0, \sigma^2),$$

then:

$$\theta + Y \sim N(\theta, \sigma^2).$$

Thus:

$$\hat{\theta} = \theta + \frac{Z_n}{\sqrt{n}} \approx N\left(\theta, \frac{1}{nI(\theta)}\right).$$

Final Result

$$\hat{\theta} \approx N\left(\theta, \frac{1}{nI(\theta)}\right)$$

5.7 Derivation Idea (Outline)

1. Start with the score function:

$$U(\theta) = \frac{d}{d\theta} \ell(\theta).$$

2. At the MLE:

$$U(\hat{\theta}) = 0.$$

3. Use a Taylor expansion around θ :

$$U(\hat{\theta}) \approx U(\theta) + (\hat{\theta} - \theta) \ell''(\theta).$$

4. Rearranging:

$$\hat{\theta} - \theta \approx -\frac{U(\theta)}{\ell''(\theta)}.$$

5. Use:

- $U(\theta) \sim N(0, nI(\theta))$
- $\ell''(\theta) \approx -nI(\theta)$

6. Result:

$$\hat{\theta} \approx N\left(\theta, \frac{1}{nI(\theta)}\right).$$

Example 5.7.1

Bernoulli Case

Let:

$$X_1, X_2, \dots, X_n \sim \text{Bernoulli}(\theta),$$

where:

$$P(X_i = x_i) = \theta^{x_i} (1 - \theta)^{1-x_i}.$$

Let:

$$k = \sum_{i=1}^n X_i.$$

Step 1: Log-Likelihood

$$\ell(\theta) = \log L(\theta) = k \log \theta + (n - k) \log(1 - \theta).$$

Step 2: Score Function

$$U(\theta) = \frac{d}{d\theta} \ell(\theta) = \frac{k}{\theta} - \frac{n - k}{1 - \theta}.$$

Step 3: MLE Condition At the MLE:

$$U(\hat{\theta}) = 0.$$

Solve:

$$\frac{k}{\hat{\theta}} = \frac{n - k}{1 - \hat{\theta}}.$$

$$\hat{\theta} = \frac{k}{n} = \bar{X}.$$

Step 4: Second Derivative

$$\ell''(\theta) = \frac{d^2}{d\theta^2} \ell(\theta) = -\frac{k}{\theta^2} - \frac{n - k}{(1 - \theta)^2}.$$

Step 5: Taylor Expansion We expand the score function around θ :

$$U(\hat{\theta}) \approx U(\theta) + (\hat{\theta} - \theta) \ell''(\theta).$$

But:

$$U(\hat{\theta}) = 0,$$

so:

$$0 \approx U(\theta) + (\hat{\theta} - \theta) \ell''(\theta).$$

Step 6: Rearranging

$$\hat{\theta} - \theta \approx -\frac{U(\theta)}{\ell''(\theta)}.$$

Step 7: Distribution of $U(\theta)$ Recall:

$$k = \sum_{i=1}^n X_i.$$

$$\mathbb{E}[k] = n\theta, \quad \text{Var}(k) = n\theta(1 - \theta).$$

Thus:

$$U(\theta) = \frac{k}{\theta} - \frac{n - k}{1 - \theta}.$$

Rewrite:

$$U(\theta) = \frac{k - n\theta}{\theta(1 - \theta)}.$$

Using the Central Limit Theorem:

$$k - n\theta \approx N(0, n\theta(1 - \theta)).$$

Therefore:

$$U(\theta) \approx N\left(0, \frac{n}{\theta(1 - \theta)}\right).$$

Step 8: Approximate $\ell''(\theta)$

Take expectation:

$$\mathbb{E}[\ell''(\theta)] = -\frac{n\theta}{\theta^2} - \frac{n(1 - \theta)}{(1 - \theta)^2} = -\frac{n}{\theta(1 - \theta)}.$$

Thus:

$$\ell''(\theta) \approx -\frac{n}{\theta(1 - \theta)}.$$

Step 9: Substitute into Approximation

$$\hat{\theta} - \theta \approx -\frac{U(\theta)}{\ell''(\theta)}.$$

Substitute:

$$\hat{\theta} - \theta \approx -\frac{U(\theta)}{-\frac{n}{\theta(1 - \theta)}} = \frac{\theta(1 - \theta)}{n}U(\theta).$$

Step 10: Distribution of $\hat{\theta}$

Since:

$$U(\theta) \approx N\left(0, \frac{n}{\theta(1-\theta)}\right),$$

multiply by $\frac{\theta(1-\theta)}{n}$:

$$\hat{\theta} - \theta \approx N\left(0, \frac{\theta(1-\theta)}{n}\right).$$

Thus:

$$\boxed{\hat{\theta} \approx N\left(\theta, \frac{\theta(1-\theta)}{n}\right)}$$

Final Interpretation

- The MLE $\hat{\theta} = \bar{X}$ is approximately normal for large n .
- Mean: θ
- Variance: $\frac{\theta(1-\theta)}{n}$
- Larger $n \Rightarrow$ smaller variance $\Rightarrow$ more precise estimate.

Example 5.7.2

Normal Distribution (Mean, Variance Known)

Let:

$$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2),$$

where σ^2 is known and μ is unknown.

Step 1: Write the Log-Likelihood The density is:

$$f(x_i; \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right).$$

Thus:

$$\ell(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Step 2: Score Function Differentiate:

$$U(\mu) = \frac{d}{d\mu} \ell(\mu).$$

$$U(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu).$$

Step 3: MLE Condition Set:

$$U(\hat{\mu}) = 0.$$

$$\sum (x_i - \hat{\mu}) = 0 \Rightarrow \hat{\mu} = \bar{X}.$$

Step 4: Second Derivative

$$\ell''(\mu) = \frac{d^2}{d\mu^2} \ell(\mu) = -\frac{n}{\sigma^2}.$$

Step 5: Taylor Expansion Expand the score around μ :

$$U(\hat{\mu}) \approx U(\mu) + (\hat{\mu} - \mu)\ell''(\mu).$$

Since:

$$U(\hat{\mu}) = 0,$$

we get:

$$0 \approx U(\mu) + (\hat{\mu} - \mu)\ell''(\mu).$$

Step 6: Rearranging

$$\hat{\mu} - \mu \approx -\frac{U(\mu)}{\ell''(\mu)}.$$

Step 7: Distribution of $U(\mu)$ Recall:

$$U(\mu) = \frac{1}{\sigma^2} \sum (X_i - \mu).$$

Now:

$$\sum (X_i - \mu) \sim N(0, n\sigma^2),$$

because sums of normal variables are normal.

Thus:

$$U(\mu) \sim N\left(0, \frac{n\sigma^2}{\sigma^4}\right) = N\left(0, \frac{n}{\sigma^2}\right).$$

Step 8: Substitute $\ell''(\mu)$

$$\ell''(\mu) = -\frac{n}{\sigma^2}.$$

Thus:

$$\hat{\mu} - \mu \approx -\frac{U(\mu)}{-\frac{n}{\sigma^2}} = \frac{\sigma^2}{n} U(\mu).$$

Step 9: Final Distribution Since:

$$U(\mu) \sim N\left(0, \frac{n}{\sigma^2}\right),$$

multiply by $\frac{\sigma^2}{n}$:

$$\hat{\mu} - \mu \sim N\left(0, \frac{\sigma^2}{n}\right).$$

Thus:

$$\boxed{\hat{\mu} \sim N\left(\mu, \frac{\sigma^2}{n}\right)}$$

Step 10: Fisher Information

$$I(\mu) = \frac{1}{\sigma^2} \quad \Rightarrow \quad I_n(\mu) = \frac{n}{\sigma^2}.$$

Thus:

$$\frac{1}{nI(\mu)} = \frac{\sigma^2}{n}.$$

Final Result

$$\boxed{\hat{\mu} \approx N\left(\mu, \frac{1}{nI(\mu)}\right)}$$

Interpretation

- The MLE $\hat{\mu} = \bar{X}$ is exactly normal (not just asymptotically).
- Mean: μ
- Variance: $\frac{\sigma^2}{n}$
- Larger sample size $\Rightarrow$ more precise estimate.